CS616 Project Proposal: Hybrid Simulated Annealing + k-opt for TSP*

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I. Project Title

Solving The TSP Problem: A Simulated Annealing With K-opt Operators.

II. Student Information

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TABLE I
Student Information and Justification

III. Problem Statement

The Traveling Salesman Problem (TSP) is a well-known combinatorial optimization problem that seeks to find the shortest possible route visiting each city exactly once and returning to the starting point. The objective is to minimize the total travel cost while ensuring the validity of a single Hamiltonian tour [1]. The TSP can be formulated as follows:

$$\text{Minimize} \quad f(x) = \sum_{i=1}^{n} \sum_{\substack{j=1 \\ j \neq i}}^{n} d_{ij} x_{ij} \tag{1}$$

$$\text{subject to} \quad \sum_{\substack{j=1 \\ j \neq i}}^{n} x_{ij} = 1, \quad \forall i = 1, 2, \ldots, n \tag{2}$$

$$\sum_{\substack{i=1 \\ i \neq j}}^{n} x_{ij} = 1, \quad \forall j = 1, 2, \ldots, n \tag{3}$$

$$\sum_{i \in Q} \sum_{\substack{j \in Q \\ j \neq i}} x_{ij} \leq |Q| - 1, \quad \forall Q \subsetneq \{1, \ldots, n\}, |Q| \geq 2 \tag{4}$$

$$x_{ij} \in \{0, 1\}, \quad \forall i, j = 1, 2, \ldots, n \tag{5}$$

Constraint (2) ensures that each city has exactly one outgoing edge, while constraint (3) ensures that each city has exactly one incoming edge. Constraint (4) eliminates subtours, enforcing the connectivity of all cities into a single Hamiltonian cycle.

A. Relevance

The TSP is considered NP-hard and serves as a fundamental benchmark for evaluating the performance of metaheuristic algorithms such as Simulated Annealing (SA), Genetic Algorithms (GA), Particle swarm optimization (PSO), and Ant Colony Optimization (ACO). Its computational complexity and wide range of real-world applications (e.g., logistics, routing, circuit design) make it a suitable test problem for optimization research.

IV. Literature Review

Soler and de Lambertye propose a fast GRASP metaheuristic for the Trigger Arc TSP, a dynamic variant where arc costs depend on previously traversed arcs. Their approach combines a novel MIP-based construction heuristic with multi-neighborhood local search, significantly outperforming exact MIP solvers can on large-scale instances. [2] Costa et al. present a Self-Cooling Simulated Annealing (SCSA) algorithm that adaptively adjusts cooling schedules for nonlinear least-squares optimization. The method incorporates parameter separation and dynamic Gaussian sampling, demonstrating superior performance over gradient-based and standard Monte Carlo approaches in multi-modal landscapes. While effectively balancing exploration and convergence, the algorithm requires problem-dependent tuning of thermal parameters for optimal performance [3].

V. Proposed Algorithm

In this project, rather than solving the program exactly, a hybrid metaheuristic approach based on the *Simulated Annealing (SA)* algorithm combined with the *k-opt* neighborhood structure is developed to obtain high-quality approximate solutions efficiently. Additional techniques such as *reheating* and *perturbation* are used to improve exploration and avoid local minima.

A. Algorithm Overview

The Simulated Annealing (SA) algorithm is inspired by the physical process of annealing in metallurgy, where a material is slowly cooled to reach a low-energy crystalline state. In optimization, this concept is translated into a probabilistic search mechanism that allows occasional acceptance of worse solutions to escape local minima.

SA is integrated with the local search operator k-opt to refine candidate tours by iteratively replacing k edges to obtain improved paths. Reheating dynamically increases the temperature when stagnation is detected, while perturbation introduces controlled randomness to diversify the search and promote exploration.

B. Implementation Details

The algorithm is implemented in Python using the `tsplib95` library [4] to load benchmark datasets from the TSPLIB repository. The initial solution is generated using greedy heuristics. Temperature control follows an exponential cooling schedule, while acceptance probability is controlled by the Metropolis criterion [5].

1) Metropolis Acceptance Criterion

General Form [5]

$$P(\text{accept}) = \min\left(1, \exp\left(\frac{\Delta E}{T}\right)\right) \tag{6}$$

Expanded Form [5]

$$P(\text{accept } s'|s, T) = \begin{cases} 1 & \text{if } f(s') \le f(s) \\ \exp\left(\frac{f(s)-f(s')}{T}\right) & \text{if } f(s') > f(s) \end{cases} \tag{7}$$

2) Temperature Schedule

Exponential cooling [6]

$$T_k = T_0 \cdot \alpha \quad \text{where} \quad \alpha \in (0,1) \tag{8}$$

Initial temperature [6]

$$T_0 = C \cdot \bar{\Delta E} \quad \text{or} \quad T_0 \ge \max_{\text{observed}} |\Delta E| \tag{9}$$

Markov chain length [6]

$$M = \beta \cdot n \tag{10}$$

3) Reheating

Progressive reset [6]

$$T^{(i)} = \frac{T_0}{i}, \quad \text{where} \quad i = 1, 2, \ldots, N_{\text{reheat}} \tag{11}$$

4) k-opt Local Search

2-opt distance change [7]

$$\Delta d = d(c_i, c_j) + d(c_{i+1}, c_{j+1}) - d(c_i, c_{i+1}) - d(c_j, c_{j+1}) \tag{12}$$

k-opt neighborhood size [7]

$$|N_k(s)| = O(n^k) \tag{13}$$

Neighborhood function [6]

$$N(s) = \{s' : s' \text{ is obtained from } s \text{ by one move}\} \tag{14}$$

5) Perturbation

Adaptive perturbation strength [8]

$$k = k_{\min} + \left\lfloor (k_{\max} - k_{\min}) \cdot \frac{T}{T_0} \right\rfloor \tag{15}$$

6) Stopping Criteria

General stopping criteria [5]

$$\text{Stop if } T \le T_{\min} \quad \text{or} \quad \text{iterations} \ge K_{\max} \tag{16}$$

VI. Expected Results and Evaluation

The proposed hybrid *Simulated Annealing + k-opt* algorithm will be evaluated on several Traveling Salesman Problem (TSP) instances of varying sizes:

A. Dataset

Selected TSPLIP95 instances for experimentation as follows

- Small-scale: 2 instances, <50 cities
- Medium-scale: 4 instances, 300–500 cities
- Large-scale: 4 instances, <1000 cities

B. Metrics

- Tour length (total distance)
- Optimality gap (%)
- Computational time (CPU time)
- Convergence behavior
- Standard deviation (stability)
- Performance plots

C. Baseline Methods for Comparison

- Nearest Neighbor (NN)
- Simulated Annealing (SA)
- Local Search
- SA + k-opt (proposed)
- Gurobi (exact solver)

D. Parameter Configuration

Parameter	Value / Range	Description
T_0	$0.1d$	Initial temperature controlling exploration intensity
$T_{\min}$	$10^{-3}T_0$	Minimum temperature; stopping limit
α	0.95–0.99	Cooling rate; slower rate gives higher-quality solutions
β	10–100	Markov chain multiplier determining iteration length
$k_{\min}$	2	Minimum perturbation strength (2-opt neighborhood)
$k_{\max}$	5	Maximum perturbation strength (up to 5-opt)
$K_{\max}$	10^3–10^4	Maximum number of outer iterations; scales with n
N_{reheat}	3–5	Number of reheating cycles for diversification

TABLE II
SIMULATED ANNEALING + K-OPT PARAMETER CONFIGURATION.

VII. Expected Contributions

The expected contributions of this project are as follows.

- Design of a hybrid optimization approach that combines exploration through Simulated Annealing with focused local refinement using k-opt moves, enabling more effective navigation of complex solution spaces in the Traveling Salesman Problem.

References

[1] G. Reinelt, "Tsplib—a traveling salesman problem library," *ORSA Journal on Computing*, vol. 3, no. 4, pp. 376–384, 1991.

[2] J. S. Soler and G. de Lambertye, "A fast grasp metaheuristic for the trigger arc tsp with mip-based construction and multi-neighborhood local search," 2025.

[3] F. Costa, G. Quintero, M. Riddick, L. Andrini, A. Vahnovan, and C. Huck Iriart, "Self-cooling simulated annealing (scsa) algorithm for nonlinear least square's data analysis," *Journal of Mathematical & Computer Applications*, vol. 4, pp. 2–6, 08 2025.

[4] R. H. Grant, "tsplib95: A python library for loading and handling tsplib instances," https://tsplib95.readthedocs.io/, 2018–2025, accessed: 2025-10-30.

[5] S. Kirkpatrick, C. D. Gelatt, and M. P. Vecchi, "Optimization by simulated annealing," *Science*, vol. 220, no. 4598, pp. 671–680, 1983.

[6] E.-G. Talbi, *Metaheuristics: From Design to Implementation*. Wiley, 2009.

[7] K. Helsgaun, "An effective implementation of k-opt moves for the lin-kernighan tsp heuristic," Roskilde University, Tech. Rep., 2009.

[8] H. R. Lourenço, O. C. Martin, and T. Stützle, "Iterated local search," in *Handbook of Metaheuristics*, F. Glover and G. Kochenberger, Eds. Springer, 2010, pp. 363–397.